

Submission Summary

Conference Name

2nd International Conference on Intelligent Computing and Automation for Sustainable Solutions

Track Name

ICASS2027

Paper ID

135

Paper Title

Comparative analysis of Machine Learning and deep learning approaches for handwritten digit recognition using the MNIST dataset

Abstract

Handwritten digit recognition is a foundational problem in computer vision and pattern recognition, with widespread applications in postal automation, banking, and document digitization. This paper presents a systematic comparative study of five machine learning and deep learning algorithms applied to digit classification using the MNIST benchmark dataset (70,000 grayscale images, 28×28 pixels): Convolutional Neural Networks (CNN), Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forest Classifiers, and K-Nearest Neighbors (KNN). The proposed CNN architecture achieved a classification accuracy of 99.1%, outperforming all other approaches. Advanced techniques including online data augmentation, Adam optimization, batch normalization, and dropout regularization contributed to the model's generalization. A detailed comparative analysis covering accuracy, inference speed, memory requirements, and model size is presented, alongside a systematic error analysis and practical deployment guidelines for production environments.

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CNNs, Machine Learning and Data Analysis

Submission Files

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